

The Chain Rule & Directional Derivatives

ANSWER KEY



The multivariable chain rule

- 1 Let $z = x^2 + y^2$, $x = \cos t$, $y = \sin t$. Find $\frac{dz}{dt}$.
 $\frac{\partial z}{\partial x} = 2x$, $\frac{\partial z}{\partial y} = 2y$; $\frac{dx}{dt} = -\sin t$, $\frac{dy}{dt} = \cos t$. $\frac{dz}{dt} = 2x(-\sin t) + 2y\cos t = -2\cos t \sin t + 2\sin t \cos t = 0$
(consistent with $z = \cos^2 t + \sin^2 t = 1$, a constant).
- 2 Let $w = xy + z^2$, $x = t$, $y = t^2$, $z = t^3$. Find $\frac{dw}{dt}$ at $t = 1$.
 $\frac{\partial w}{\partial x} = y$, $\frac{\partial w}{\partial y} = x$, $\frac{\partial w}{\partial z} = 2z$. $\frac{dw}{dt} = y(1) + x(2t) + 2z(3t^2)$. At $t = 1$: $x = y = z = 1$, so $\frac{dw}{dt} = 1 + 2 + 6 = 9$.

Chain rule with two independent variables

- 3 Let $z = f(x, y)$ where $x = s + t$, $y = s - t$. Express $\frac{\partial z}{\partial s}$ in terms of $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.
 $\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s} = \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y}$.
- 4 For $z = x^2y$, $x = s + t$, $y = s - t$, find $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t}$ at $s = 2$, $t = 1$.
At $s = 2$, $t = 1$: $x = 3$, $y = 1$. $\frac{\partial z}{\partial x} = 2xy = 6$, $\frac{\partial z}{\partial y} = x^2 = 9$. Since $\frac{\partial x}{\partial s} = 1$, $\frac{\partial y}{\partial s} = 1$: $\frac{\partial z}{\partial s} = 6 + 9 = 15$. Since $\frac{\partial x}{\partial t} = 1$, $\frac{\partial y}{\partial t} = -1$: $\frac{\partial z}{\partial t} = 6 - 9 = -3$.

Gradient and directional derivative

- 5 Find the gradient of $f(x, y) = x^2y^3$ at $(1, 2)$.
 $\nabla f = \langle 2xy^3, 3x^2y^2 \rangle$. At $(1, 2)$: $\langle 2(8), 3(4) \rangle = \langle 16, 12 \rangle$.
- 6 Find the directional derivative of $f(x, y) = x^2y$ at $(2, 1)$ in the direction of $\mathbf{v} = \langle 1, 1 \rangle$.
 $\nabla f = \langle 2xy, x^2 \rangle$; at $(2, 1)$: $\langle 4, 4 \rangle$. Unit vector $\hat{\mathbf{v}} = \langle \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \rangle$. $D_{\mathbf{v}}f = \frac{4}{\sqrt{2}} + \frac{4}{\sqrt{2}} = \frac{8}{\sqrt{2}} = 4\sqrt{2}$.

The gradient points in the direction of steepest ascent

- 7 For $f(x, y) = 3x^2 - y^2$ at $(1, 2)$, find the direction of steepest ascent and the maximum rate of increase.
 $\nabla f = \langle 6x, -2y \rangle$; at $(1, 2)$: $\langle 6, -4 \rangle$. Direction of steepest ascent: $\langle 6, -4 \rangle$. Maximum rate of increase:
 $|\nabla f| = \sqrt{36 + 16} = \sqrt{52} = 2\sqrt{13}$.
- 8 In what direction from $(1, 1)$ does $f(x, y) = xy$ decrease most rapidly, and at what rate?
 $\nabla f = \langle y, x \rangle$; at $(1, 1)$: $\langle 1, 1 \rangle$. Steepest decrease is $-\nabla f = \langle -1, -1 \rangle$, at rate $|\nabla f| = \sqrt{2}$.

Application of the gradient

- 9 The temperature on a plate is $T(x, y) = 100 - x^2 - 2y^2$. A bug at $(2, 1)$ wants to cool off fastest. In which direction should it move?
- $\nabla T = \langle -2x, -4y \rangle$; at $(2, 1)$: $\langle -4, -4 \rangle$ (direction of steepest warming). To cool fastest, move in $\langle 4, 4 \rangle$ (or the unit direction $\langle \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \rangle$).
- 10 Find the rate of change of $T(x, y) = 100 - x^2 - 2y^2$ at $(2, 1)$ in the direction of the unit vector $\langle \frac{3}{5}, \frac{4}{5} \rangle$.
- $\nabla T(2, 1) = \langle -4, -4 \rangle$ (from above). $D_{\mathbf{v}}T = \langle -4, -4 \rangle \cdot \langle \frac{3}{5}, \frac{4}{5} \rangle = -\frac{12}{5} - \frac{16}{5} = -\frac{28}{5}$.